Some answers came fast, some others took weeks, if not months, of probing and prodding.

One such question that remained unanswered till the end was the unusually high tolerance band of Viking's gas generator temperature. For a gas generator with an operating pressure of 600 degrees Celsius, the tolerance band of plus-or-minus 50 degrees Celsius was unusual, we thought. In engineering, an ideal product should have zero tolerance, which means it works as designed, to the point. In this case, the high temperature tolerance of the gas chamber meant fluctuations through other systems including the pump and combustion chamber.

We asked SEP engineers why the tolerance band was set so high.

'Can't it be around plus-or-minus 10 degrees Celsius?'

There was no answer.

Later we understood that Arianne Space, too, had raised this question to SEP, which said it was not worth studying and correcting the tolerance band since it would delay the project and incur higher costs. Since it was generally accepted that the situation did not pose a real threat to the system, it was left at that.

We noted down all the points that we did not understand and discussed. We documented all these discussions and sent them regularly to a backup team of about twenty engineers in India, headed by T P Rangamani. The backup team fired back questions at us, and this served as a double brainstorm.

Sending these documents was not easy. I requested the Indian embassy in Paris to allot a diplomatic bag for us to send the documents to India. A diplomatic bag was immune to scanning or that is what we were made to understand. The embassy wanted to know what was so secretive about the documents that came from our workplace.

'There was nothing secretive,' I reasoned, 'but these are meant only for our scientists.' Contradictory it might have sounded, but ambassador R D Sathe agreed. Still there was a problem. The diplomatic bags were dispatched to India from Paris only once a week, so I have to store the documents for six days before depositing them at the embassy.